

Automated Multi-Stage Sleep Classification Using Wrist-Worn Photoplethysmography Signal

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Abstract— Sleep plays a vital role in human life. Classifying sleep stages is important for measuring sleep quality as well as detecting any sleep disorders. Traditional sleep staging requires a complex medical setup; several sensors are used to record polysomnography signals, which is both expensive to collect and challenging for participants for long term monitoring, as it may disrupt their sleep. In this study, we classified different sleep stages, including Wake vs. Sleep, Wake vs. Non-Rapid Eye Movement (NREM) vs Rapid Eye Movement (REM), and Wake vs. Light Sleep vs. Deep Sleep vs. REM. We utilized data from 10 participants in the DREAMT dataset, extracting 48 different features from the BVP (PPG) signal across five feature categories: waveform morphology, respiratory modulation, heart rate variability, statistical and energy measures, and complexity and fractal dynamics. We classified the data into two-stage, three-stage, and four-stage categories using the Random Forest Algorithm, k-nearest neighbors algorithm, and Extreme Gradient Boosting (XGBoost) algorithm. Among these classifiers, the XGBoost classifier outperformed. The classification results for the two-class, three-class, and four-class problems using the XGBoost algorithm were 86.35%, 78.20%, and 75.20%, respectively. These findings demonstrate that wearable PPG-based sleep staging can provide a reliable, low-cost, and non-intrusive alternative to traditional polysomnography for sleep monitoring.

Keywords— Photoplethysmography, Multi-class sleep stage, Classification, Machine Learning, XG-Boost

I. INTRODUCTION

Sleep stage classification has shifted from traditional polysomnography scoring to automatic systems, which is capable of analysing physiological signals through computational models. The photoplethysmography (PPG) signal has become one of the most practical alternatives because it is cost effective, lightweight, and it can be easily integrated into wearable devices. There are five stages of sleep, where four of them are non-rapid eye movement (NREM) [1]. EEG, EOG, and ECG signals can capture physiological activities, such as EEG reflects brain waves, EOG tracks eye movements, and ECG records cardiac rhythms. Which allows machine learning models to distinguish different sleep stages such as REM, NREM, and wake with greater precision [2].

Automated sleep stage classification using cardiorespiratory signals, specifically electrocardiogram (ECG) and thoracic

respiratory effort, which is a practical alternative to conventional polysomnography-based methods [3]. Bozkurt et al. proposed that single-channel ECG signals can be effectively used for sleep–wake detection by deriving heart rate variability and extracting QRS components across multiple frequency bands. The ECG-derived features were then analysed and classified using hybrid machine learning models to accurately distinguish between sleep and wake stages [4]. Instantaneous heart rate derived from ECG signal was used in fully convolutional network which is based on dilated convolutions to capture both local and long-range dependencies, shows comparable performance to PSG-based models [5].

Sleep stage classification using wrist-worn PPG and accelerometer signal, which aim to provide a comfortable, low-cost, and scalable alternative to traditional ECG or PSG-based systems [6]. Motin et al. proposed a model using PPG extracted statistical parameters such as mean, variance, skewness, kurtosis, and signal energy and classified them using support vector machines (SVM) and KNN for binary sleep–wake detection [7]. Signal quality was improved by applying band-pass and Savitzky–Golay filters and dividing the data into 30-second epochs before computing time- and frequency-domain HRV and spectral features. Using these features, sleep–wake classification is conducted using SVM classifier [8]. A multivariate logistic regression model with LASSO feature selection is developed to identify sleep and wake states from PPG signals. These features are related to heart rate, respiration, and movement, which are used to classify sleep and wake states from smartphone-based pulse oximeter data [9]. Using Random Forest (RF) classifiers using statistical, temporal, and nonlinear PPG features, together with SHAP-based interpretation, showed accuracy of 77.8 % and 69.2 % for three- and four-class classification and improved interpretability for wearable applications [10]. Habib et al. developed a convolutional neural network (CNN) model that can automatically learn and extract relevant features directly from the raw PPG signal for multi-stage sleep classification [11]. Wearable systems employing PPG-based ensemble tree models and SVM classifiers reported 76–78 % accuracy for multi-stage sleep detection, confirming the feasibility of PPG as a reliable bio-signal for continuous sleep monitoring [12]. Although PPG-based sleep staging has

been studied before, most works focus on binary classification and limited features, highlighting the need for a comprehensive analysis using diverse PPG features for robust multi-class sleep stage classification. In this study, we aimed to classify multi-class sleep stages using a diverse PPG feature set encompassing waveform morphology, respiratory modulation, HRV, statistical energy, and nonlinear complexity.

II. MATERIALS AND METHOD

A. Dataset Overview

In this study, we use publicly available DREAMT dataset [13]. The data were recorded using Empatica E4 wristband and the Dreem 2 headgear, which recorded multimodal physiological data. The Blood Volume Pulse (BVP), which is also known as PPG is sampled at a rate of 64 HZ. The dataset also includes accelerometer, electrodermal activity, skin temperature, heart rate (HR), and inter beat interval channels, along with reference sleep stage annotations. The PPG signals were annotated indirectly by aligning them with expert sleep technician-scored PSG sleep stages for each 30-second epoch. In this work, we have used data of ten subjects to train and validated our proposed model.

B. Data Preprocessing

The PPG signal was split into 30 second epochs to correspond with the conventional sleep stage assessment technique. After 30 seconds of segmentation, a Savitzky-Golay finite impulse response filter with a window length of nine samples and a third-order polynomial is implemented to smooth short-term fluctuations and reduce motion-related noise while preserving the waveform's natural shape. The cleaned and preprocessed epoch is used during feature extraction. The detail of the proposed model is illustrated in Fig. 1.

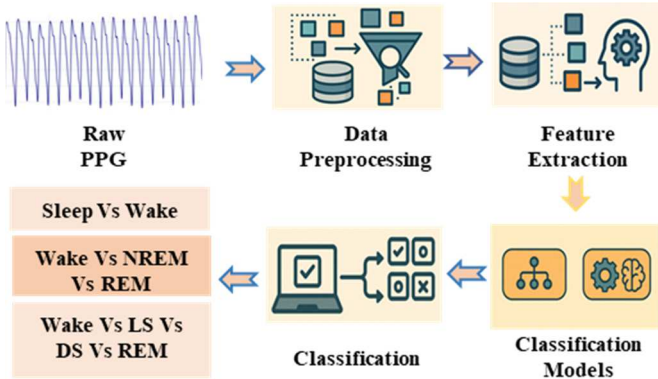


Fig. 1. Block Diagram of Multiple Sleep Stage Classification Workflow

C. Feature Extraction

We extracted 48 features from five different categories such as: Various morphological, respiratory modulation, statistical and energy, heart rate variability and complexity, and fractal dynamics from PPG signals, as indicated in TABLE I. The morphological features are the mean total area beneath the PPG waveform, systolic and diastolic areas, systolic and diastolic time intervals, inflection point area ratio, and inflection point time ratio. We also obtained the standard deviations for all of

these features. We also extract temporal data including pulse-to-pulse intervals, respiration-induced amplitude modulation, and frequency modulation features. We removed temporal variables such as variations in pulse-to-pulse interval and respiration-related changes in signal amplitude and frequency to examine how heart rate and breathing patterns altered during different stages of sleep. Heart rate and its variability fluctuate during various sleep stages due to alterations in autonomic nervous system activity. PPG-based features like pulse-to-pulse interval (PPI), root mean square of successive differences (RMSSD), and amplitude variations can identify these changes. For determine the effect of complex physiological dynamics, a number of non-linear and entropy-based features such as Hjorth parameters, fractal dimensions were extracted, which can detect irregularities and self-similar patterns associated with transitions in autonomic activity. Energy and shape-related features, such as curve length and Teager energy, were also extracted to detect oscillatory behaviour and morphological variations of the signal. Hjorth parameter can detect the signal's variance, frequency characteristics, and dynamic changes, respectively

TABLE I. Features Extracted from PPG Signal.

Features Category	Features	Description of Features
Waveform Morphology Features	Mean of A, A1, A2, T1, T2, IPAR, IPTR Standard Deviation (SD) of A, A1, A2, T1, T2, IPAR, IPTR	A1: Systolic Area, A2: Diastolic Area, A: Area Under PPG, T1: Systolic Time, T2: Diastolic Time, IPAR: Inflection point area ratio, IPTR: Inflection point time
Rhythm and Respiratory Modulation Features	Mean of PPI, RIAM, RIFM, SD of PPI, RIAM, RIFM, SD of successive differences of PPI, RIAM, RIFM, Root mean square of successive differences of PPI, RIAM, RIFM	PPG peak to peak interval (PPI), Respiration-induced amplitude modulation (RIAM), Respiration-induced frequency modulation (RIFM).
Statistical and Energy Features	avgADPPG, stdADPPG, MADPPG, IQRPPG, nCMPG, avgEPPG, SFPPG, avgCLPPG, avgTEPPG, GMPPG, HaMPPG, TM25PPG, TM50PPG, SkewPPG, KurtPPG	Mean absolute deviation, SD of absolute deviation, Median absolute deviation of the PPG signal, Interquartile range of PPG signal, nth central moment of the PPG signal, Average energy of PPG signal, Shape factor, Average curve length, Average Teager energy, Geometric, Harmonic mean, 25% trimmed mean, 50% trimmed mean, Skewness of PPG signal.
Heart rate Variability Features	SD1PPG, SD2PPG, RSD1SD2PPG, CCMPPG	Poincaré SD1, Poincaré SD2 of PPG, Ratio of SD1 to SD2 from the Poincaré plot, Complex Correlation Measure (CCM) of the PPG signal.
Complexity and Fractal Dynamics Features	HAPPG, HMPPG, HCPPG, HDFPPG, LCPPG, KFDPPG	Hjorth activity, Hjorth mobility, Hjorth complexity, Higuchi fractal dimension, Lyapunov exponent, Katz fractal dimension of the PPG signal.

D. Classification Algorithms

In this study, three different machine learning algorithms: Random Forest, k-Nearest Neighbors (KNN), and Extreme Gradient Boosting (XGBoost) were employed to address the multi-stage sleep classification task. The classification was performed at multiple levels, including binary classification

TABLE II. Various Performance value obtained for Two-Class, Three-Class, Four- Class Classification

	2 Class Classification				3 Class Classification					4 Class Classification					
	Avg Acc	Acc (Wake)	Acc (Sleep)	F1 Score	Avg Acc	Acc (Wake)	Acc (NREM)	Acc (REM)	F1 Score	Avg Acc	Acc (Wake)	Acc (LS)	Acc (DS)	Acc (REM)	F1 Score
Random Forest	84.50	50.52	95.16	83.30	76.19	55.35	91.82	25.30	74.01	71.25	57.85	88.25	46.55	23.49	69.10
KNN	81.19	49.74	91.05	80.35	74.81	57.18	87.63	34.34	73.67	67.25	54.45	80.24	56.03	31.33	66.22
XG-Boost	83.75	54.71	92.86	83.00	78.25	60.05	89.91	46.39	77.40	74.62	61.52	87.39	55.17	46.39	73.75

(Sleep vs. Wake), three-class classification (Wake vs. NREM vs. REM), and four-class classification (Wake vs. Light Sleep vs. Deep Sleep vs. REM).

The Random Forest algorithm is a machine learning classifier which employs a number of decision trees constructed from random subsets of data and features. From these decision trees, multiple decision will be built. Using bagging process this algorithm will train data. This classifier is effective for analyzing biomedical signal as it can deal with noisy, high-dimensional data. This work implements the Random Forest classifier with a maximum tree depth of 15 and 200 estimators. KNN algorithm is a classification method that uses distance for decision-making. We used Euclidean distance with five neighbors. The missing data were filled using the median value, and all features were normalized using standard scaling. The XGBoost technique constructs a number of decision trees. The XGBoost algorithm is trained using 200 boosting rounds, with each tree having a maximum depth of 8. The model used “mlogloss” as its evaluation metric, which is suitable for handling multiple sleep stage classes. Label encoding was turned off to simplify processing, and a fixed random state was used to keep the results consistent.

To evaluate the performance of the proposed models, five key performance metrics were employed: accuracy, sensitivity, specificity, precision, and F1-score. All 30-second epochs were combined and split using a stratified 80/20 training/testing split to preserve sleep stage proportions, with missing values median-imputed and all features standardized before training.

III. RESULT AND DISCUSSION

In the case of two-class classification, the Random Forest, KNN, and XGBoost algorithms achieve 84.50%, 81.19%, and 83.75% accuracy, respectively. The classifiers achieved 76.19%, 74.81%, and 78.25% accuracy for the three class. The accuracy of the same classifiers for four-class classification is 71.25%, 67.25%, and 74.62%. TABLE II presents the performance comparison of Random Forest, KNN, and XGBoost classifiers for classifying two-class, three-class, and four-class sleep stage. The XGBoost method consistently perform well between the three classifiers. The confusion matrices of the XGBoost classifier for two, three, and four class sleep stage classifications is demonstrated in Fig. 3.

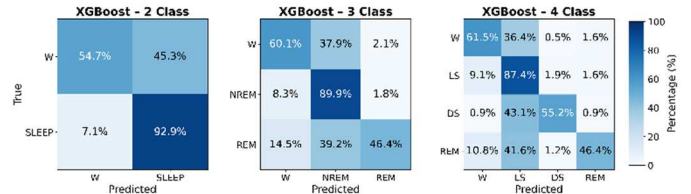


Fig. 2. Confusion matrix of XGBoost Classifier for a) two class, b) three class and c) four class classification.

The classification accuracy obtained using each PPG based feature categories for two-class, three-class, and four-class models is shown in TABLE III. Among all feature categories, statistical and energy features deliver the highest accuracy, followed by complexity and fractal dynamics features while heart rate variability features perform the weakest.

TABLE III. Classification Accuracy of Different PPG-based Feature Categories for 2-Class, 3-Class, and 4-Class Models.

Features Category	Accuracy (2-Class)	Accuracy (3-Class)	Accuracy (4-Class)
Waveform morphology	78.75 %	69.12 %	64.69 %
Rhythm and respiratory modulation	79.12 %	68.88 %	62.06 %
Statistical and energy	82.81 %	75.88 %	70.44 %
Heart rate variability	72.12 %	60.19 %	51.94 %
Complexity and fractal dynamics	81.44 %	74.56 %	69.19 %

The significance of each feature was analyzed using the LIME framework to provide interpretability of the model’s decision process. Among the 48 extracted features, TM25PPG, LCPPG, KurtPPG, SkewPPG, and Mean_A2 were identified as the top five contributors. The top fifteen most significant features along with their corresponding importance values are presented in Fig. 4. These features effectively capture the underlying physiological variations associated with different sleep stages. Specifically, TM25PPG and LCPPG indicate waveform variability and nonlinear dynamics of the photoplethysmography (PPG) signal, while KurtPPG and SkewPPG characterize morphological asymmetries and shape alterations linked to heart rhythm transitions. The morphological feature Mean_A2 reflects changes in pulse amplitude and diastolic components, which are influenced by blood pressure fluctuations and vascular tone modulation across various sleep stages.

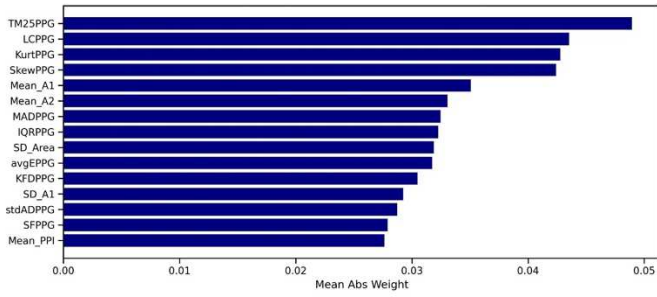


Fig. 3. Lime global feature importance for 4 class classification using XGBoost.

The impact of increasing the number of significant features on the performance of the proposed model is illustrated in Fig. 5, which depicts the variation in classification accuracy with respect to the number of selected features. The accuracy exhibits a sharp increase up to approximately 15 features, after which it tends to stabilize across all classification tasks. The highest classification accuracies achieved were 86.30%, 79.60%, and 75.00% for two-, three-, and four-class classifications, using 15, 20, and 30 features, respectively. Furthermore, the hypnogram comparison between the actual and predicted sleep stages, presented in Fig. 6, corroborates the reliability and robustness of the proposed method.

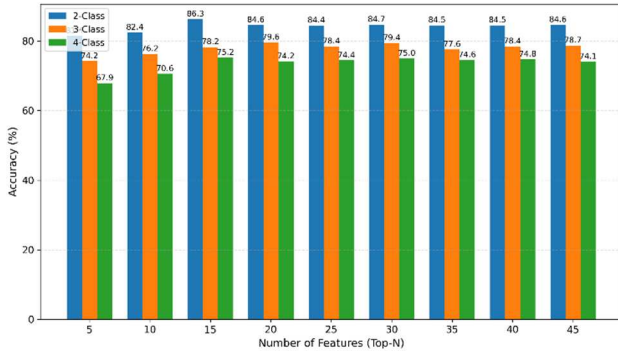


Fig. 4. Accuracy vs. Number of Selected Features for 2-Class, 3-Class, and 4-Class Classification.

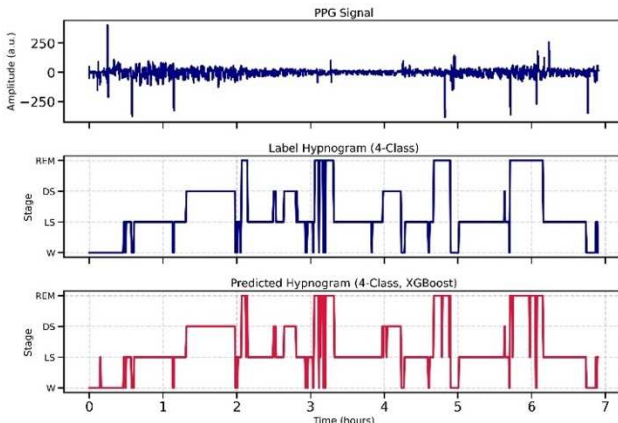


Fig. 5. Comparison between the actual sleep stage hypnogram and predicted sleep stage hypnogram for four class sleep stage classification of a participant.

Table IV. summarizes studies using PPG-based features for sleep stage classification. Previous works employed methods

like KNN, CNN, Random Forest, Least Absolute Shrinkage and Selection Operator (LASSO), SVM etc. The proposed model achieved higher and more consistent performance across 2-, 3-, and 4-class classifications, which demonstrates its robustness in detecting sleep stages from PPG signals.

TABLE IV. Summary of Studies Conducted Based on PPG-Based Features.

Study	Method	Classes	Accuracy (%)
Habib et al. [11]	CNN	2	94.40
F. Smarandache et al. [10]	RF	2, 3, 4	82.56, 77.79, 69.20
Dekhordi et al. [9]	LASSO	2	77
Motin et al. [12]	SVM	2, 3, 4	84.66, 79.62, 72.23
Proposed	XGBoost	2, 3, 4	86.35, 78.20, 75.20

IV. CONCLUSION

This study demonstrates that PPG signals derived from wearable devices provide a simple, low-cost, and comfortable alternative for sleep stage monitoring compared to traditional multi-sensor clinical setups. Among the evaluated classical machine learning algorithms, XGBoost achieved the highest accuracy for two, three and four class sleep stage classification. Feature analysis revealed that using the top 15 features offers an optimal balance between simplicity and performance, highlighting the potential for efficient models with fewer features. Category-wise examination of PPG features further elucidates their discriminative significance to identify different sleep stages. Future work will extend this approach to five-class sleep classification and explore deep learning-based methods.

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